IRON WITHIN RESEARCH

Thymosin Alpha 1 — Certificates of Analysis

Every batch we have tested for this product, newest first.
Match the strength on your vial to the rows below and turn to that page.

PAGE	STRENGTH	BATCH / LOT
2–3	10mg, 42mg	10/9/2026 · Lot IWR-1010-TA1
4–8	2mg, 10mg, 96mg	7/24/2026 · BATCH-0063
9	5mg, 41mg	5/6/2026

Page numbers include this index page. Testing is per production batch; one batch can cover several vial sizes.

SUMMIT
BIOANALYTICS

Certificate of Analysis

#SBA-26-0302

Summit BioAnalytics | Nassau, The Bahamas | lab@summitbioanalytics.com | summitbioanalytics.com

TESTED FOR
Iron Within Research

SAMPLE
Thymosin Alpha 1 — 10 mg

PASS

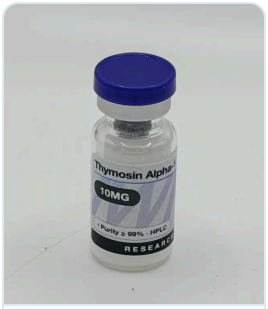
BATCH #	PURITY	MASS	IDENTITY
IWR-1010-TA1	99.52%	10.42 mg	Confirmed

COA #:	SBA-26-0302	Method:	Full QC Panel
Lot Number:	IWR-1010-TA1	Analysis Date:	2026-10-09
Labeled Content:	10 mg	Date Received:	2026-10-06
		Appearance:	White lyophilized powder, intact seal

SCAN TO VERIFY

- ✓ Identity
- ✓ Purity
- ✓ Endotoxins
- ✓ Fentanyl Free

Test	Specification	Result
Identity	Thymosin Alpha 1	Confirmed
Purity (HPLC)	>= 98.0%	99.52%
Content (per vial)	Report only	10.42 mg/vial
Fentanyl Screen (LC-MS/MS)	Not detected	Not detected
Endotoxins (LAL)	Report Result	0.02 EU/mL
Sterility Screen (rapid PCR)	No growth	No growth
Heavy Metals (ICP-MS)	ICH Q3D limits	All analytes within limits



SAMPLE AS RECEIVED

Qualitative and quantitative chemical analysis by RP-HPLC with UV detection; identity confirmed by LC-MS. Fentanyl screening by targeted LC-MS/MS. Sterility by rapid nucleic-acid amplification — a screening method, not compendial sterility per USP <71>. Full methodology available on request.

Kasey Malganius

Kasey Malganius
Lab Director
2026-10-09

COA #:	**SBA-26-0302**
Access Code:	**RQX3EM27**
Analyst:	**A. Joseph**
Verify:	summitbioanalytics.com/verify
Issued:	2026-10-09

This certificate relates only to the item tested, as received. Research use only — not for human or veterinary use. A not-detected screening result describes this sample only and is not a statement about any other unit or lot.

Horizon Analytical

Certificate of Analysis

Lot Number: DPS-8269267
Brand / Distributed by: Iron Within Research
Identity: www.ironwithinresearch.com

Analysis Date: 07/24/2026
Searchable via: horizonanalytical.com

Compound:	Thymosin Alpha-1
Report Number:	1020365
Appearance:	White Lyophilized Powder

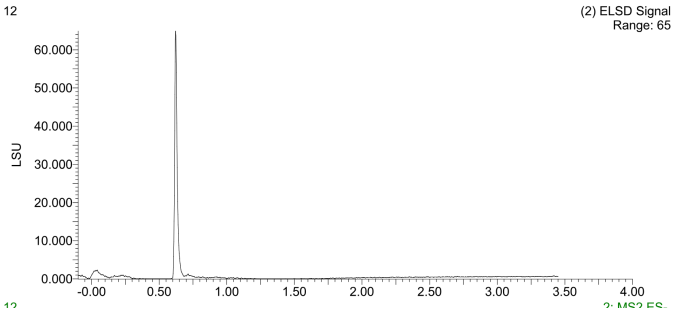
CAS:	62304-98-7
Formula:	$C_{129}H_{215}N_{33}O_{55}$
Mol Weight:	~3108.3 g/mol

Pubchem CID: 16130571

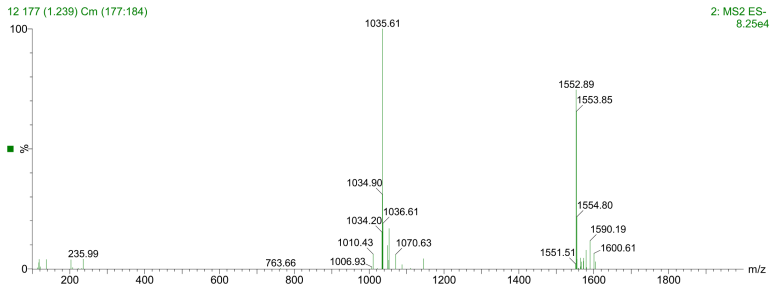
Qualitative and Quantitative chemical analysis by Ultra High Performance Liquid Chromatography with Mass Spectrometry

	Specification	Result	Scan to Validate:
Compound Test:	Thymosin Alpha-1	Thymosin Alpha-1	
Quantity:	10mg	9.96mg	
Purity:	>98%	99.39%	

UPLC



MS



Aleksey Yevtodiyenko PhD
Research and Formulation Chemist

This purity analysis was conducted using UPLC/MS under standard laboratory conditions, following validated analytical protocols to ensure accurate and reliable results. This analysis is intended for informational and research applications.

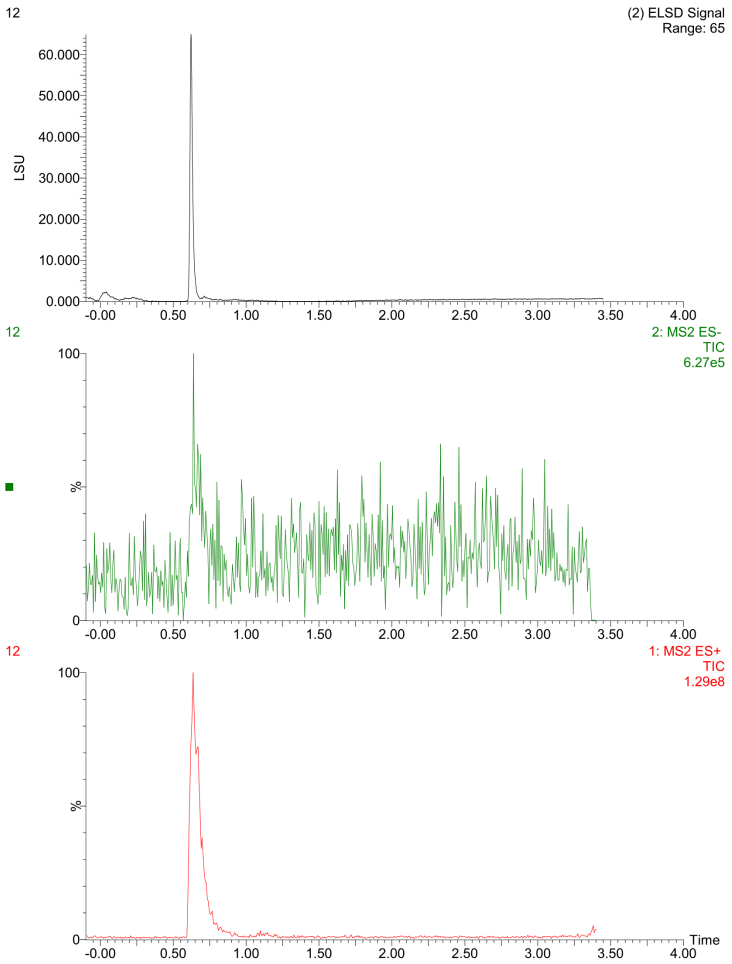
Horizon Analytical

Certificate of Analysis

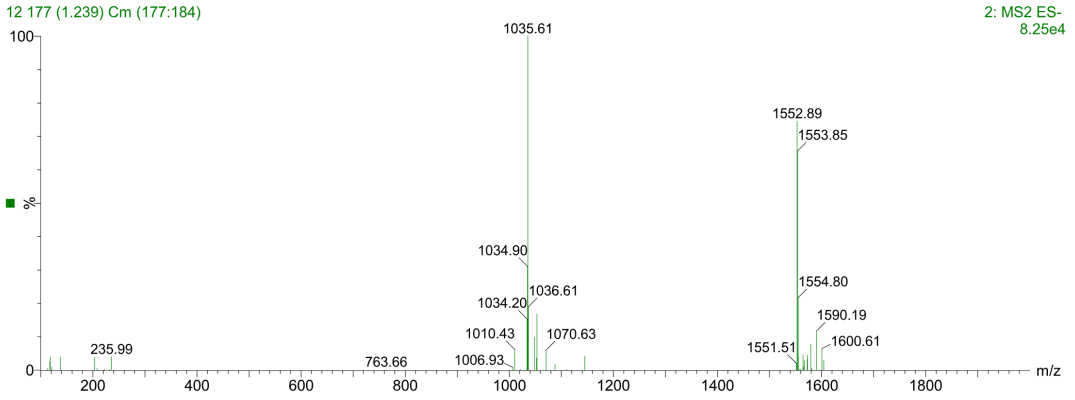
Lot Number: DPS-8269267
Brand / Distributed by: Iron Within Research
Identity: www.ironwithinresearch.com

Analysis Date: 07/24/2026
Searchable via: horizonanalytical.com

Thymosin Alpha-1 (10mg) • Pubchem CID: 16130571
Ultra High Performance Liquid Chromatography (UPLC)



Mass Spectrometry (MS)



Horizon Analytical

Certificate of Analysis

Lot Number: DPS-8269267
Brand / Distributed by: Iron Within Research
Identity: www.ironwithinresearch.com

Analysis Date: 07/24/2026
Searchable via: horizonanalytical.com

Compound:	Thymosin Alpha-1
Report Number:	1020364
Appearance:	-

CAS:	62304-98-7
Formula:	$C_{129}H_{215}N_{33}O_{55}$
Mol Weight:	~3108.3 g/mol

Pubchem CID: 16130571

Endotoxin Test

	Specification	Result	Scan to Validate:
Compound Test:	Thymosin Alpha-1	Pass	
Endotoxin:	-	< 0.05 EU/mL	

Aleksey Yevtodiyenko PhD
Research and Formulation Chemist

This endotoxin analysis was performed under standard laboratory conditions using validated testing methodologies to ensure accurate and reliable results. The analysis is intended for informational and research purposes only.

Contact at: contact@horizonanalytical.com

Proudly Owned and Operated in the USA

SUMMIT
BIOANALYTICS

Certificate of Analysis

#SBA-26-0063

Summit BioAnalytics | Nassau, The Bahamas | lab@summitbioanalytics.com | summitbioanalytics.com

TESTED FOR
Iron Within Research

SAMPLE
Thymosin Alpha 1 — 10 mg

PASS

BATCH #	PURITY	MASS	IDENTITY
IWR-2607-TA110	99.54%	10.02 mg	Confirmed

COA #:	SBA-26-0063	Method:	Full QC Panel
Lot Number:	IWR-2607-TA110	Analysis Date:	2026-08-02
Labeled Content:	10 mg	Date Received:	2026-07-29
		Appearance:	White lyophilized powder, intact seal

SCAN TO VERIFY

- Identity
- Purity
- Endotoxins
- Fentanyl Free

VIAL CAP
Blue
#0017A8

CLOSURE AS RECEIVED

Test	Specification	Result
Identity	Thymosin Alpha 1	Confirmed
Purity (HPLC)	>= 98.0%	99.54%
Content (gross)	Report only	10.02 mg
Fentanyl Screen (LC-MS/MS)	Not detected	Not detected
Endotoxins (LAL)	Report Result	0.03 EU/mL

Test	Specification	Result
Sterility Screen (rapid PCR)	No growth	**No growth**
Heavy Metals (ICP-MS)	ICH Q3D limits	**All analytes within limits**

Qualitative and quantitative chemical analysis by RP-HPLC with UV detection; identity confirmed by LC-MS. Fentanyl screening by targeted LC-MS/MS. Sterility by rapid nucleic-acid amplification — a screening method, not compendial sterility per USP <71>. Full methodology available on request.



Kasey Malganius
Lab Director
2026-08-02

COA #:	**SBA-26-0063**
Access Code:	**5JUEQ972**
Analyst:	**R. Whitfield**
Verify:	summitbioanalytics.com/verify
Issued:	2026-08-02

This certificate relates only to the item tested, as received. Research use only — not for human or veterinary use. A not-detected screening result describes this sample only and is not a statement about any other unit or lot.

FREEDOM DIAGNOSTICS

Certificate of Analysis

FreedomDiagnosticsTesting.com

Admin@FreedomDiagnostics.net



Client:	Iron Within Research
Accession #:	2605060159
Search Code:	Iron2605060159
Received:	05/06/2026
Reported:	05/08/2026
Lot:	Light Blue Cap

Sample Summary

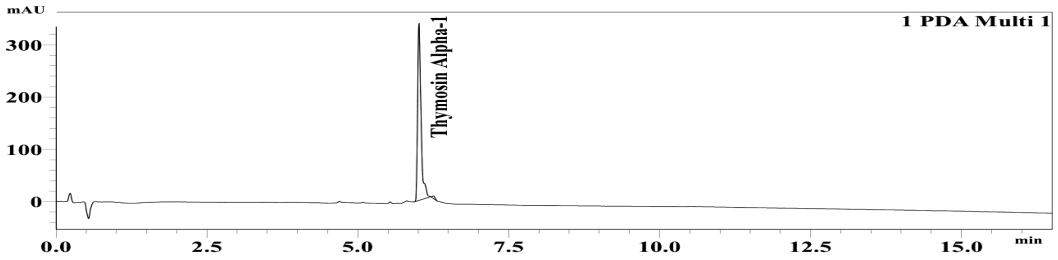
Product:	Thymosin Alpha 5mg	**Purity:**	99.00%
Identity:	Confirmed	**Net Content:**	3.41 mg
Appearance:	White Lyophilized Powder		

Analytical Results

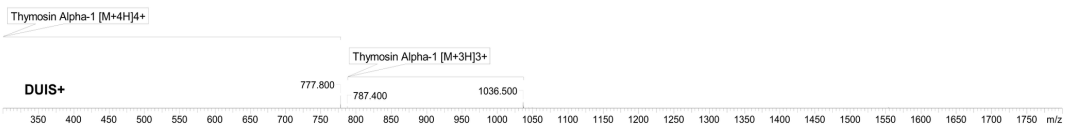
Test	Result
Identity (LC-MS)	Thymosin Alpha-1
Purity (HPLC-UV)	99.00%
Net Content	3.41 mg

Method: HPLC with UV detection coupled with mass spectrometry (LC-MS).

Chromatogram



Mass Confirmation



Alex Johnson

Principal Chemist

FreedomDiagnosticsTesting.com

Admin@FreedomDiagnostics.net

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The peptide purity analysis reported here was conducted using LCMS/MS under standard laboratory conditions. This analysis is intended for informational purposes only and is specific to the sample(s) provided. The peptides tested are intended for research use only and are not approved for human or veterinary use, diagnostic, therapeutic, or clinical applications. Results should be interpreted by qualified professionals within the scope of the intended research. The accuracy and reliability of the test may be influenced by sample integrity, handling, and other experimental variables.